Complete schematic - sheet 5

Every component terminal is shown. Equal net labels are connected; use connection IDs in the CSV for physical routing.

E1 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 02 Nov 2026

R25 | ECHO upper resistor
ADDITIONAL / MISSING OR SELECT
2.2k 1%, >=1/8 W
1 — EL_RAW
2 — EL_SAFE

R26 | ECHO lower resistor
ADDITIONAL / MISSING OR SELECT
3.3k 1%, >=1/8 W
1 — EL_SAFE
2 — GND

R27 | ECHO upper resistor
ADDITIONAL / MISSING OR SELECT
2.2k 1%, >=1/8 W
1 — ER_RAW
2 — ER_SAFE

R28 | ECHO lower resistor
ADDITIONAL / MISSING OR SELECT
3.3k 1%, >=1/8 W
1 — ER_SAFE
2 — GND

R1 | L_FWD input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — L_FWD_3V3
2 — GND

R2 | L_REV input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — L_REV_3V3
2 — GND

R3 | R_FWD input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — R_FWD_3V3
2 — GND

R4 | R_REV input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — R_REV_3V3
2 — GND

R5 | ENABLE input pull-down
ADDITIONAL / MISSING OR SELECT
2.2k 1%, >=1/8 W
1 — ENABLE_3V3
2 — GND

R6 | TF input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — TF_3V3
2 — GND

R7 | TL input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — TL_3V3
2 — GND

R8 | TR input pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — TR_3V3
2 — GND

R9 | BRUSH_IN1 pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — BRUSH_IN1
2 — GND

R10 | BRUSH_IN2 pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — BRUSH_IN2
2 — GND

R11 | CONVEYOR_IN3 pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — CONVEYOR_IN3
2 — GND

R12 | CONVEYOR_IN4 pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — CONVEYOR_IN4
2 — GND

R13 | DRV_L.R_EN local EN pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — EN_DRIVER_5V
2 — GND

R14 | DRV_L.L_EN local EN pull-down
ADDITIONAL / MISSING OR SELECT
10k 1%, >=1/8 W
1 — EN_DRIVER_5V
2 — GND

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.